

Basic Concepts and Economic Principles for Decision Making

Module 1

Management 405

Prof. Nico Voigtländer · UCLA Anderson

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Module 1 · Introduction

Introduction

▪ About the instructor

- BA in Economics and Environmental Engineering in Berlin/Germany
- MSc in Environmental Engineering and Technology Policy from MIT
- PhD in Economics in Barcelona
- Visiting positions at Harvard, University College London

▪ My research: Why are some countries so rich and others so poor?

▪ Contact information

- Email: nico.v@ucla.edu

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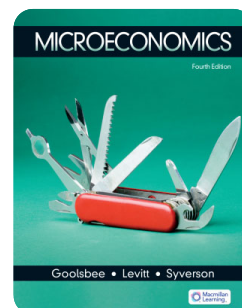
Logistics I

- TA for the class:
 - Rafael Rubio: TA405.EMBA2@gmail.com
 - PhD student in Economics & Management
- Course website: <https://nvoigtla.github.io/MGMT-405-EMBA/>
 - All course materials posted, including class handouts, slides
 - (Optional) Math Review
 - Videos
 - Classes recorded + livestreamed (Links on BruinLearn)

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Logistics II

- Textbook: Goolsbee, Levitt, Syverson (4th edition)
- Weekly readings/podcasts
 - Send me suggestions!
- Breaks during class:
 - Approx. every 1 – 1.5 hours



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Logistics III

- Overall **5** Problem Sets
 - Due dates on the Class Calendar (on the class website)
 - Study groups pre-assigned
 - BruinLearn Dropbox or via email to TA
 - Questions on problem sets: Email TA
 - AI use allowed for studying
- Further logistics in Course Syllabus
 - Attendance, participation, grading
 - Midterm 35%, Final 40%, Problem Sets 25%
- Midterm Exam/Final Exam: **AI tools not allowed during exams**

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Questions and Office Hours

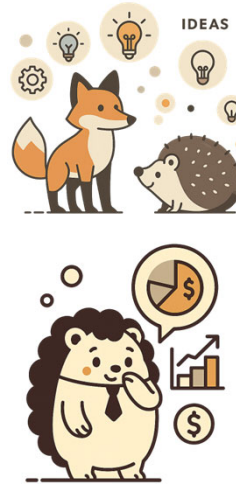
- **TA responsible for** questions related to:
 - Problem sets, practice exercises, practice exams
- Prof. **Nico**: All **conceptual** questions (topics from class)
- “Coffee & Econ” chats via Zoom with Nico & TA
 - Biweekly on weekends (time TBA). Tentative next session: **Sunday, Oct 11 at 9am Pacific Time.**
 - Zoom link on BruinLearn
 - Talk about concepts from class and applications, then questions about problem sets
- Additional Online TA review sessions each week (recorded)
 - TA will send out doodle poll

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Module 1 · Introduction

Recall from Video 1: Economists as Hedgehogs

- “The fox knows many things, but the hedgehog knows one big thing” (Greek aphorism).
- **Foxes** have **different strategies** for different problems. They are comfortable with nuance.
- **Hedgehogs**, on the other hand, focus on the big picture. They reduce every problem to **one organizing principle**.



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Module 1 · Introduction

Making the Most of the Course

- My main challenge: you have different backgrounds, interests, skills...
 - Some of you are Econ majors, others have not taken Econ since high school
 - Some have stronger quant backgrounds than others



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Making the Most of the Course

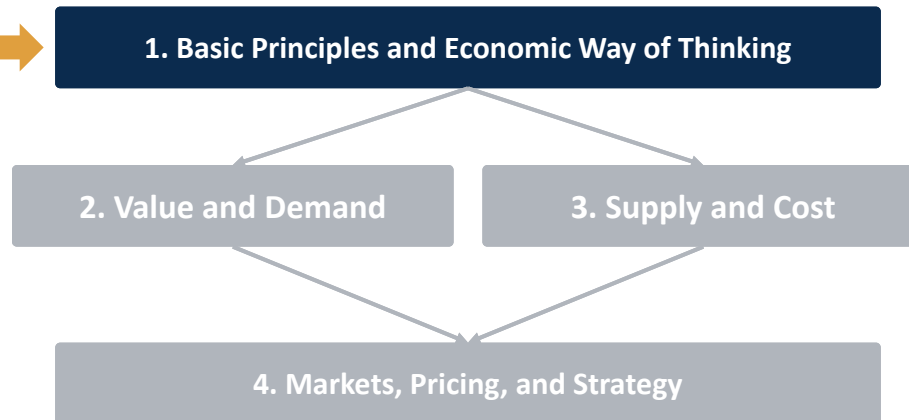
- Focus on the intuition
 - To become an economic hedgehog, you don't need to be a mathematician
 - Math is fairly simple. Do the **Math Test** and **Math Review on the class website!**
 - I will always explain the *economic intuition* behind the math
- Learn from & with your peers
 - Also: Actively use AI tools to deepen your insights. Challenge AI!
 - Again: You will not be allowed to use AI in exams

Teaching Philosophy

- Economics takes a lot of critical thinking:
 - I don't care about memorizing formulas
 - The real challenge is to figure out the right tools (and to do so, you need to build economic intuition)
- **YOU** are the expert in your field:
 - I can't solve the problems you face.
 - But I can help you ask the right questions.

Agenda for the Class

we are here →



▶ Can these prices be optimal?

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Homo Economicus



Homo Economicus (seeks to maximize utility or happiness)

- 100% selfish
- 100% rational
- 100% informed



Real Human

- Some are altruistic
- Some make stupid decisions
- Some are poorly informed

Podcast: Who is Homo Economicus? ▶

▶ Backup: Does money buy happiness?

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Module 1 · Agenda

Outline of Module 1

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Introduction

Video 1: a brief introduction of the instructor and the class

2

Markets

Video 2: what a market is, and how far it extends

3

Demand and Supply

Video 3: how buyers and sellers each respond to price

4

Equilibrium

Video 4: where demand meets supply, and what moves it

5

Economic Costs Include Opportunity Costs

In class: the value of the next-best alternative you gave up

6

Ignore Sunk Costs

In class: money already spent should not drive the next decision

7

Use Cost-Benefit and Marginal Analysis

In class: compare the benefit of the next unit with its cost

Video 1

Video 2

Video 3

Video 4

In class

In class

In class

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Module 1

In-Class Part

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Some Applications of the Material Covered in Videos 2 to 4

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Recall from Video 2: Market Definition

Extent of a market – A Simple Test

- Which products belong to a market?
- Ask: If the price of another product changes, will demand for your product change?
- Relevant for antitrust litigation (in mergers & acquisitions)
- Important dimension: How much substitution is “enough” in court?
 - Are butter and margarine in the same market?

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Market Definition Mini-Case: Kroger–Albertsons

- Oct 2022: Kroger agrees to buy Albertsons for \$24.6B — the largest US supermarket merger ever proposed
- Everything would hinge on **how the court would define the market**:
 - The firms' argument: "we are small next to **Walmart, Amazon, Costco** "
 - FTC argument: "the market is **"supermarkets"** — assessed city by city, where #1 is trying to buy #2"



Photos: Wikimedia Commons

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Is a "Costco Run" a Substitute for Everyday Shopping?

Court decision:

THE MARKET = "supermarkets" — Kroger · Albertsons · Safeway ...

"A monthly trip to Costco to stock up ... does not make a 'Costco run' a reasonable substitute for a weekly one-stop visit to a supermarket."
— Judge Adrienne Nelson, D. Or. (Dec 2024)

Outside the court's market definition:

Club stores — Costco, Sam's Club

Limited assortment — Aldi, Trader Joe's

Dollar & convenience stores

Online-only sellers



Photo: Wikimedia Commons

Dec 2024: federal and state courts block the deal — "some substitution is not enough substitution"

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Market Definition: The Case of Netflix

Define Netflix's market

- Online streaming?
- All shows/movies, incl. theatres?
- All entertainment?



Class Discussion

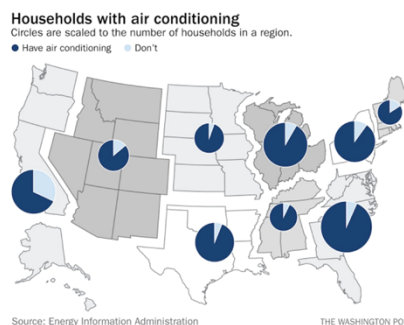
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How Can Heatwaves Affect the Demand for ACs?

Los Angeles Times

CALIFORNIA

Brutal heat wave breaks records across Southern California. When will it ease up?



Poll Break

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What is the likely effect of heatwaves on demand for AC?

- The demand curve shifts to the left 0%
- A movement up along the (same) demand curve 0%
- The demand curve shifts right 0%
- A movement down along the (same) demand curve 0%
- None of the above 0%

POLL

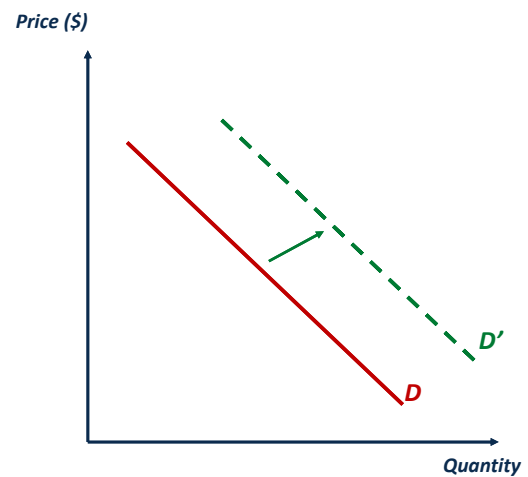
Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app

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Module 1 · In Class · Examples · Supply and Demand

Shifts of the Demand Curve for AC

The heatwaves should cause the demand curve to **shift outward (to the right)**



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Swiftonomics

THE WALL STREET JOURNAL.



- How did Taylor Swift's engagement affect the demand for diamonds?

Poll Break

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How does the decline in engagements affect the demand for diamonds?

- | | |
|---|----|
| The demand curve shifts to the left | 0% |
| A movement up along the (same) demand curve | 0% |
| The demand curve shifts right | 0% |
| A movement down along the (same) demand curve | 0% |
| None of the above | 0% |

POLL

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Shortages When Disasters Loom



- Why are there frequently shortages when major disasters loom (storms, the Covid pandemic...)?
- What role does the perception of 'fairness' play in this context?

Class Discussion

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Steps to Analyze "Shocks" to Supply and Demand

1. Figure out what the shock is
2. Determine whether the shock shifts the demand or supply curve
3. Draw the market's demand and supply curves before and after the shock

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Example: How Did the War in Ukraine Affect the Price of Wheat?



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Mini-Case: Los Angeles Residential Real Estate

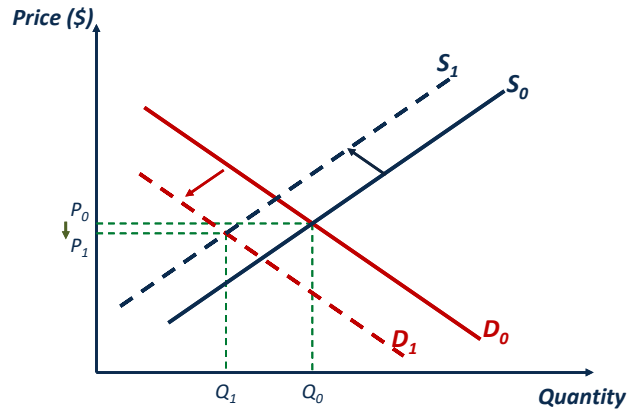


- From Sep-2021 to Sep-2025, in Los Angeles (Redfin data):
 - Inflation-adjusted home prices: ▼5%
 - Home sales: ▼40%
- Explain using supply and demand framework!

Class Discussion

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Market for Los Angeles Residential Real Estate



- **Demand curve:** shift inward due to higher mortgage rates and out-migration
- **Supply curve:** owners "locked in" to lower mortgage rates and postponed selling

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Outline of Module 1

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In class: the value of the next-best alternative you gave up | In class |
| 6 | Ignore Sunk Costs | In class |
| 7 | Use Cost-Benefit and Marginal Analysis | In class |

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Module 1 · Economic Costs Include Opportunity Costs

Opportunity Costs



Principles of Economics
Core Principle 2 - The Opportunity Cost Principle
Think Like An Economist

- The opportunity cost is the **value** of the next best alternative sacrificed
 - what you give up to get something
- The opportunity cost is not always monetary. It is typically **implicit** rather than explicit
 - Explicit cost: actual payment
 - Implicit cost: value of forgone alternative action

Full economic cost = implicit cost + explicit cost

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Module 1 · Economic Costs Include Opportunity Costs

Opportunity Cost: A Simple Example

Suppose that I can choose one of the following:



Alternatives	My Value	My Opp. Cost
Banana	9	5
Apple	5	9
Cucumber	3	9
Dragon fruit	1	9



- The **opportunity cost** is the **value** of the next best alternative foregone (the lost *opportunity*)
- Opportunity cost depends on **preferences** – it differs across individuals

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Module 1 · Economic Costs Include Opportunity Costs

Example: Opportunity Costs When Buying a Present

- 3 possible presents, price of each is \$50
- But different values to recipient



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Module 1 · Economic Costs Include Opportunity Costs

What Is the Full Economic Cost of an MBA Degree?

- Distinguish explicit vs. implicit
-
-
-
-
-
- Do remote options change the (opportunity) cost?

Class Discussion

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Module 1 · Economic Costs Include Opportunity Costs

Flip a House or Work?

- Flip house:
 - 1 year of full-time work
 - Buy for 500k
 - Invest another 100k (+ your full time)
 - Expected sale price in a year: 700k
- Consulting job offer:
 - Full-time job (not compatible with flipping the house)
 - Pays 150K
- What is the **economic** profit of flipping the house?

 Problem Set 1

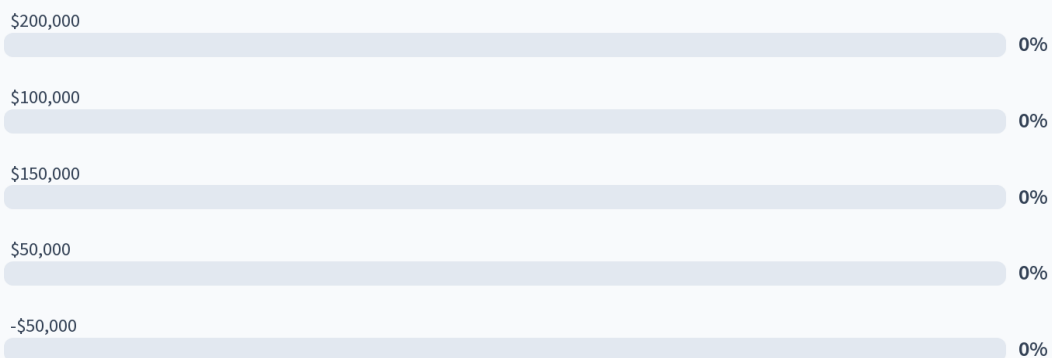
Poll Break

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What is the economic profit of flipping the house?

 0


POLL

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Another Possible Opportunity Cost

- Additionally:
 - What about the opportunity cost from the 500K of investment? That could also add to opportunity cost.
 - [Podcast](#) on opportunity costs of buying a house (will potentially discussed during our next Coffee&Econ):

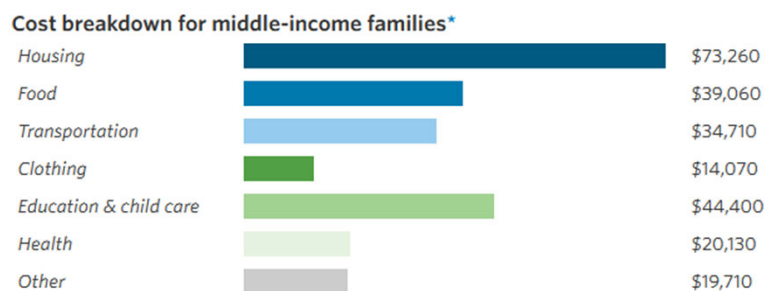
Should You Rent or Buy? The New Math.

For many millennials, buying a home is almost entirely out of reach.

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The (Full) Cost of Raising a Child

- Data from the U.S. Department of Agriculture: Costs of raising a child born in the US in 2013, to age 18:



- Which costs are missing?

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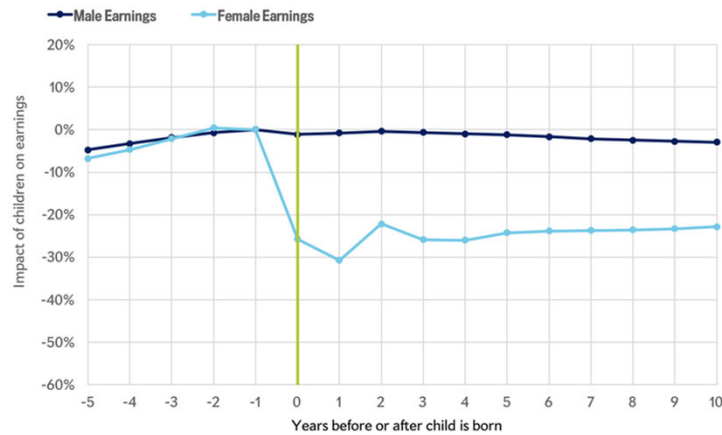
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Module 1 · Economic Costs Include Opportunity Costs

How children affect men's and women's earnings over time in Denmark

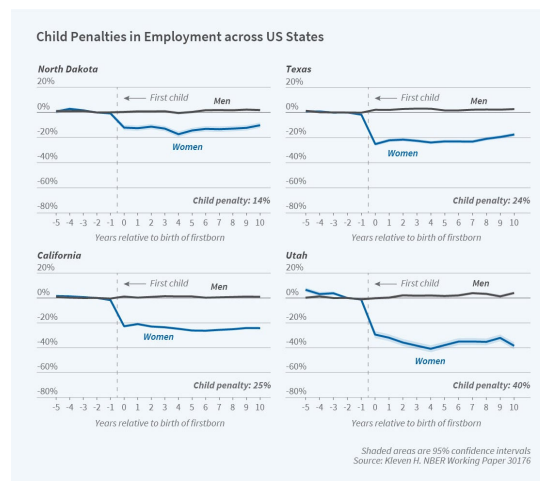
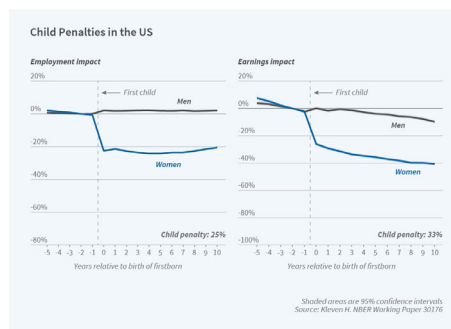
Source: Kleven, Landais, and Søgaard, "Children and Gender Inequality: Evidence from Denmark." NBER, 2018.



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Module 1 · Economic Costs Include Opportunity Costs

Similar Figures for the US, Estimated in 2022



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Module 1 · Ignore Sunk Costs

Sunk Costs

- **Sunk cost:** A cost that has been paid and cannot be recovered
- **Ignore sunk costs:** If the cost is sunk, it is not relevant for economic decisions

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Examples of Sunk Costs (You Should Ignore!)

- Individual consumers
 - Dessert in fixed-price dinners
 - Annual gym memberships (after having paid)
 - Ski tickets (after buying, return impossible)
- Firms
 - License fees (after paying)
 - Advertising (after paying)
 - To be continued in Module 3!



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The Case of Concorde

- Very coveted and salient innovation:
 - New York to London in under 3 hours
 - Pride of Air France and British Airways
- Massive financial losses
 - Never recovered investments from innovation
 - Did not even cover operating costs
- Economically rational decision would have been: shut down as soon as possible.
- Concorde eventually retired after crash in 2000



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Module 1 · Ignore Sunk Costs

Sunk Costs: Take-Away

- Make an optimal decision looking forward.
- As you cannot change the past, optimize the present/future.
- Often sunk costs come from unforeseen circumstances that change the optimal choice.
- Re-optimize without considering the past costs, if they cannot be recovered.

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Module 1 · Use Cost-Benefit and Marginal Analysis

Cost-Benefit and Marginal Analysis



Core Principle 1 - The Cost-Benefit Principle
Think Like An Economist

- Objective: **Maximize net benefits** (= total benefits – total costs)
- To maximize net benefits from any activity, **marginal analysis** is required
 - Marginal benefit (MB)**: additional benefit due to an extra unit of the activity (or an extra unit of a good/service)
 - Marginal cost (MC)**: additional cost due to an extra unit
 - Includes opportunity costs!**
- Buy / go for the extra unit if $MB > MC$

Important Rule:

- Net benefits are maximized** where $MB = MC$

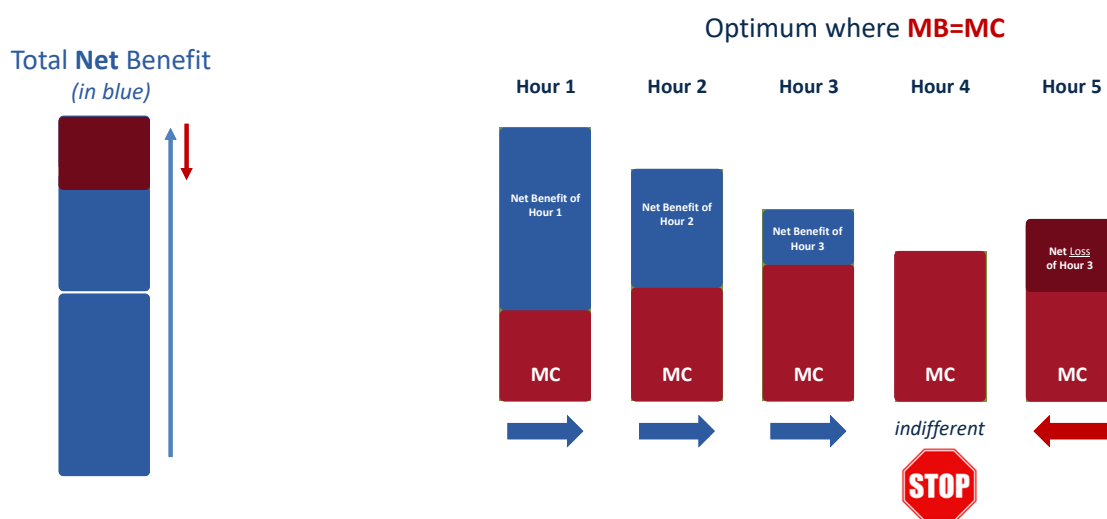
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Module 1 · Use Cost-Benefit and Marginal Analysis

Example: How Many Hours a Week Should You Exercise?



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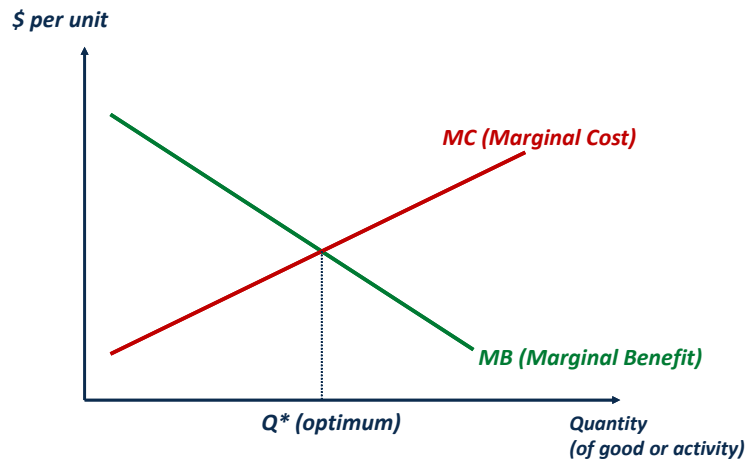
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Module 1 · Use Cost-Benefit and Marginal Analysis

Maximizing Net Benefit: The Standard Analysis That You'll See Throughout the Class

We'll generally consider the continuous case:



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Module 1 · Summary

Module 1 Summary

- Markets, supply, and demand
- Market equilibrium: intersection of supply and demand
- Key economic principles for decision-making
 - Economic costs include opportunity costs
 - Economists count implicit (opportunity) costs as well (in contrast to accountants).
 - Ignore sunk costs
 - Use cost-benefit and marginal analysis: $MB=MC$

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